SpecialtyCare · Greater Boston neuromonitoring team

IONM Team Publications

13 papers and abstracts on intraoperative neuromonitoring, with plain-language summaries. 7 are free to read in PubMed Central. Titles and links in this PDF are clickable.

Updated October 2026.

With Mass General Brigham Neurosurgery 7

Cases, techniques and reviews co-authored with neurosurgeons at Brigham and Women's Hospital and Mass General Brigham.

2026 · CASE REPORT · FREE FULL TEXT

Delayed identification and treatment of a cervicomedullary spinal vascular lesion following angiogram-negative, high-grade subarachnoid hemorrhage: illustrative case

Chang C, Motoc M, Can A, **Lachs M**, **Bose M**, Bernstock JD, Patel NJ

J Neurosurg Case Lessons. 2026;11(26):CASE26229. doi:10.3171/case26229 · PMID 42372299

A 54-year-old man with high-grade subarachnoid hemorrhage and a negative initial angiogram turned out to have an aneurysm and AVM beside the anterior spinal artery at the cervicomedullary junction. The authors argue that ventral blood distribution and spot signs in angiogram-negative SAH should prompt targeted spinal angiography at 7 to 14 days. An external ventricular drain reversed herniation and coma, and because endovascular access was restricted, the lesion was resected through a far-lateral approach with cord rotation for direct ventral exposure. D-wave and transcranial MEP monitoring told the team in real time whether a signal change was reversible ischemia or mechanical injury, and the lesion was obliterated without new deficits.

MONITORING TAKEAWAY D-wave plus transcranial MEPs separated reversible ischemic change from irreversible mechanical injury, which shaped the intraoperative response.

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2026 · SYSTEMATIC REVIEW · FREE FULL TEXT

Corticobulbar Motor Evoked Potentials: A Systematic Review of Technical Variations

Jarvis CA, **Toczylowski M**, **Bose M**, Patel RV, Wang J, McMahon JT, Bi WL

Neurosurg Pract. 2026;7(3):e000237. doi:10.1227/neuprac.0000000000000237 · PMID 42100545

A PRISMA 2020 systematic review of 72 corticobulbar MEP studies published between 2005 and 2023. Its main finding is how unsettled the technique still is: electrode montages, stimulation set-ups and the way accuracy is measured differ widely from study to study, and only a minority report sensitivity or specificity at all. Most studies used coMEPs alongside other modalities, yet few compared them directly. The authors argue that common reporting standards are what would let coMEPs spread into routine practice.

MONITORING TAKEAWAY A catalog of the coMEP montages, paradigms and warning criteria in use; useful when writing or defending a standard protocol.

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2025 · OPERATIVE VIDEO · FREE FULL TEXT

Interhemispheric Transcallosal Approach for Resection of a Pineal Region Third Ventricular to Brainstem Tumor: 2-Dimensional Operative Video

Guo X, **Bose M**, **Toczylowski M**, Fonseca A, Bi WL

Oper Neurosurg. 2025;29(6):907-908. doi:10.1227/ons.0000000000001517 · PMID 39982058

Operative video of a 19-year-old man with a pineal-region tumor reaching from the third ventricle toward the brainstem, removed through an anterior interhemispheric transcallosal, transchoroidal corridor. Subcortical mapping of the corticospinal tracts and brainstem nuclei, together with intraoperative MRI, guided the resection, and residual nodules were taken in a planned second stage to reach gross total resection. Pathology showed a mature teratoma. He improved steadily afterward and was walking on his own by eight months, with no new visual or endocrine deficits.

MONITORING TAKEAWAY Subcortical CST and brainstem-nucleus mapping combined with iMRI in a deep midline approach.

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2025 · OPERATIVE VIDEO · FREE FULL TEXT

Brain mapping during resection of high-grade brain arteriovenous malformation

Kappel AD, **Bose M**, **Toczylowski M**, Patel NJ

Neurosurg Focus Video. 2025;12(1):V7. doi:10.3171/2024.10.focvid24107 · PMID 39845307

Technique video on removing high-grade AVMs that sit in eloquent territory, here involving somatosensory cortex and the visual and corticospinal pathways. Through a bilateral craniotomy, the team stimulated cortex and subcortical white matter with a suction stimulator so the motor tracts could be followed continuously as the nidus came out, keeping roughly 3 mm from the corticospinal tract. The authors present real-time stimulation mapping as a way to make these vascular resections safer.

MONITORING TAKEAWAY Dynamic suction-stimulator mapping carried over from tumor work into AVM resection, with a ~3 mm CST margin.

Free full text (PMC) · **Publisher** · **PubMed**

2025 · OPERATIVE VIDEO · FREE FULL TEXT

Motor mapping-guided resection of a brainstem recurrent pilocytic astrocytoma

Guo X, **Bose M**, Galvin CP, Bi WL

Neurosurg Focus Video. 2025;12(1):V2. doi:10.3171/2024.10.focvid24110 · PMID 39845304

Video of a recurrent brainstem pilocytic astrocytoma removed under dynamic subcortical motor mapping with a monopolar stimulating suction device. Corticospinal responses were recorded continuously at midbrain and pontine levels, with contralateral limb and ipsilateral facial muscles captured together while tumor was removed. Resection stopped when the stimulation threshold fell below 1 mA, taken as about 1 mm or less from the tract.

MONITORING TAKEAWAY Monopolar suction-stimulator mapping with a <1 mA stop threshold and simultaneous limb and facial responses.

Free full text (PMC) · **Publisher** · **PubMed**

2025 · OPERATIVE VIDEO · FREE FULL TEXT

Multimodal-Assisted Resection of Right Insular Glioblastoma: A 2-Dimensional Operative Video

Guo X, Torio E, **Bose M**, Golby AJ, Bi WL

Oper Neurosurg. 2025;29(3):451. doi:10.1227/ons.0000000000001471 · PMID 39625292

Operative video of a 55-year-old right-handed woman with a large right insular glioblastoma extending into the frontotemporal lobes, causing brain fog and left-sided weakness, resected through a pterional transsylvian approach. The case shows a full multimodal set-up working together: gelfoam-guided intraoperative ultrasound, real-time instrument tracking on navigation, intraoperative MRI, and neuromonitoring with transcranial, corticobulbar and direct cortical MEPs, dynamic subcortical motor mapping and a brain-sag reversal strategy. The enhancing tumor was completely removed without new deficits. Pathology confirmed IDH-wildtype glioblastoma, and she began chemoradiation with better neurological function than before surgery.

MONITORING TAKEAWAY The full cranial monitoring stack in one case: transcranial, corticobulbar and direct cortical MEPs, dynamic subcortical mapping, and brain-sag reversal.

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2025 · ORIGINAL RESEARCH · FREE FULL TEXT

Interpretation and Strategy to Resolve Neuromonitoring Changes Associated With Brain Sag

Bose M, **Toczylowski M**, Guo X, Liu DD, Epplin-Zapf T, Wilent WB, Bi WL

Oper Neurosurg. 2025;28(6):841-854. doi:10.1227/ons.0000000000001430 · PMID 40372119

A case-based analysis from one senior surgeon's cranial tumor practice showing how brain sag, the brain shifting and settling during surgery, can produce false-positive losses of somatosensory and motor signals. The authors describe a recognizable pattern of stepwise signal decline and show that recovery during saline infusion indicates artifact rather than injury. They set out an interpretation and response strategy so the team can avoid changing the operation for a change that is not real.

MONITORING TAKEAWAY Pattern recognition for sag-related false-positive SSEP/MEP loss in cranial cases, with a stepwise response algorithm.

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SpecialtyCare research 6

Large outcome studies, practice guidelines and mapping techniques from SpecialtyCare's national network and partner hospitals.

2026 · GUIDELINES

Best Practice Guidelines for the Execution of Intraoperative Neuromonitoring in Spine Deformity Surgery

Anari JB, Park AM, Brown MW, Elfilali MM, Kourpas KM, Bruins BB, Franco A, Mendiratta A, Pinyavat T, Umina J, Strantzas S, Sethi R, Lewis S, Iyer RR, Menger R, Sucato DJ, Illingworth KD, Samdani AF, Lenke LG, Vitale MG

Spine Deform. 2026. doi:10.1007/s43390-026-01559-9 · PMID 42806212

Expert consensus guidelines for running IONM in routine adolescent idiopathic scoliosis and in high-risk spinal deformity surgery. A 16-member multidisciplinary panel of spine surgeons, anesthesiologists, neurologists and neurophysiologists worked through a modified Delphi process (three rounds of surveys and meetings, then a final survey, with 80% agreement required). They reached consensus on 27 items for AIS and 25 for high-risk cases, organized into IONM timeline checklists meant to standardize practice and reduce variability in care.

MONITORING TAKEAWAY Ready-made IONM timeline checklists for AIS and high-risk deformity cases.

Publisher · PubMed

2025 · ORIGINAL RESEARCH

Sensorimotor Physiological Mapping During Asleep Deep Brain Stimulation Lead Placement With Multichannel Intraoperative Neuromonitoring

Yamaguchi C, Anwar H, Lesko A, Kaber SM, Oliveria SF

Oper Neurosurg. 2025;29(3):381-388. doi:10.1227/ons.0000000000001499 · PMID 40203201

A single-surgeon series of 88 consecutive asleep deep brain stimulation cases (67 Parkinson's disease, 14 essential tremor, 7 dystonia) in which macrostimulation mapping with multichannel IONM guided lead placement alongside intraoperative CT, in place of awake testing. Targets were GPi in 60, STN in 14 and VIM in 14, and 95.5% were bilateral. Mean single-stage surgery time was 170 minutes, no leads needed revision, no unexpected sensorimotor side effects appeared at programming, and patient-reported outcomes matched awake DBS.

MONITORING TAKEAWAY IONM macrostimulation mapping gives physiological confirmation of DBS lead position under general anesthesia.

Publisher · PubMed

2023 · ORIGINAL RESEARCH

Linking Patterns of Intraoperative Neuromonitoring (IONM) Alerts to the Odds of a New Postoperative Neurological Deficit: Analysis of 27,808 Cervical Spine Procedures From a National Multi-institutional Database

Wilent WB, Tesdahl EA, Epplin-Zapf T, Cohen J, Rhee J, Klineberg EO, Harrop JS, Vaccaro AR, Sestokas AK

Clin Spine Surg. 2023;36(3):96-105. doi:10.1097/BSD.0000000000001445 · PMID 36959181

A review of 27,808 extradural cervical spine procedures (2017 to 2021) from a national multi-institutional IONM database, in patients with myelopathy, stenosis or radiculopathy. Alerts were classified by which modalities and which nerves or muscles were involved and by whether they had resolved at closure, and the odds of a new neurologic deficit were modeled while accounting for clustering by surgeon and neurophysiologist. Posterior approaches carried more risk of a new deficit than anterior ones, and both how much an alert raised the risk and how much its resolution lowered it depended on the alert's pattern.

MONITORING TAKEAWAY The pattern of an alert (which modalities, which muscles) and its status at closure carry real prognostic weight in cervical cases.

Publisher · PubMed

2021 · ORIGINAL RESEARCH

Impact of inhalational anesthetic agents on the baseline monitorability of motor evoked potentials during spine surgery: a review of 22,755 cervical and lumbar procedures

Wilent WB, Tesdahl EA, Trott JT, Tassone S, Harrop JS, Klineberg EO, Sestokas AK
Spine J. 2021;21(11):1839-1846. doi:10.1016/j.spinee.2021.07.002 · PMID 34274500

Compared baseline MEP monitorability and amplitude between total intravenous anesthesia and a balanced regimen that adds volatile inhalational agents, across 16,559 cervical and 6,196 lumbar extradural spine procedures (2017 to 2020) in a multi-institutional database. Inhalational agents significantly reduced both the chance of getting monitorable baselines and their amplitude, with the largest effect in proximal limb muscles, the ones that matter for nerve root and plexus monitoring. The authors conclude TIVA is the regimen of choice when MEPs are used.

MONITORING TAKEAWAY Volatile agents cost the most in proximal muscles; TIVA is the regimen to ask for when root and plexus MEPs matter.

Publisher · PubMed

2021 · ORIGINAL RESEARCH

Intraoperative vascular complications during 2278 cerebral endovascular procedures with multimodality IONM: relationship between signal change, complication, intervention and postoperative outcome

Wilent WB, Belyakina O, Korsgaard E, Tjoumakaris SI, Gooch MR, Jabbour P, Rosenwasser R, English JD, Kim W, Tesdahl E, Cohen J, Sestokas AK
J Neurointerv Surg. 2021;13(4):378-383. doi:10.1136/neurintsurg-2020-016604 · PMID 33443128

A review of 2,278 cerebral endovascular procedures under general anesthesia monitored with EEG and SSEPs, 763 of them also with MEPs. New motor deficits occurred in 1.2% overall: 20.4% when there was an IONM alert versus 0.09% without one. Partial signal attenuation raised the odds of a deficit sharply and complete loss raised them further, while full resolution of the alert before closure cut the odds dramatically. Adding MEPs nudged sensitivity up (92.3% vs 85.7%) at a small cost in specificity (96.7% vs 98.2%).

MONITORING TAKEAWAY Alerts from arterial complications are high-stakes, but resolving them before closure markedly lowers the risk of a deficit.

Publisher · PubMed

2020 · ORIGINAL RESEARCH

Utility of motor evoked potentials to diagnose and reduce lower extremity motor nerve root injuries during 4,386 extradural posterior lumbosacral spine procedures

Wilent WB, Tesdahl EA, Harrop JS, Welch WC, Cannestra AF, Poelstra KA, Epplin-Zapf T, Stivali T, Cohen J, Sestokas AK
Spine J. 2020;20(2):191-198. doi:10.1016/j.spinee.2019.08.013 · PMID 31479780

Tested how well MEPs detect motor nerve root injury in 4,386 extradural posterior lumbosacral procedures, where MEPs are often skipped in favor of EMG and SSEPs. Multimodality monitoring was 100% sensitive and 92.2% specific for new lower-extremity motor deficits. On their own, MEPs were far more sensitive than spontaneous EMG or posterior tibial SSEPs (100% vs 14.3% and 28.6%) and also more specific than EMG. Only 4.4% of cases had a lower-extremity MEP alert, and every one of the 14 significant new root injuries had an MEP alert still unresolved at closure.

MONITORING TAKEAWAY In lumbosacral cases, MEPs catch root injuries that EMG and SSEPs miss, without a flood of false alerts.

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